# THE KUNNETH FORMULA FOR QUASI-COMPACT QUASI-SEPARATED SCHEMES VIA SQUARE-ZERO THICKENINGS

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted span cites five cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

*Editorial note.* The proof cites no bibliography item inside the counted span.

*Allowed standard prerequisites (editorial).*

- Cohomology of a square-zero thickening and the associated exact sequences. Stacks Project, Cohomology of Schemes (cohomology of infinitesimal thickenings). Provides the exact sequences on which the comparison of the thickenings is based.
- The affine-diagonal Kunneth formula. Stacks Project, Varieties, Lemma lemma-kunneth (tag 0BKC). Handles the separated/affine-diagonal case, which the author directs the reader to for the sharper statement about the maps.

*Source reference.* Original lemma 0BED: sources/stacks-chow/tags.txt, source lines 14346–14346. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01XN: sources/stacks-chow/tags.txt, source lines 2526–2526. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01EZ: sources/stacks-chow/tags.txt, source lines 1852–1852. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 071L: sources/stacks-chow/tags.txt, source lines 8995–8995. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01FF: sources/stacks-chow/tags.txt, source lines 1868–1868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $k$ be a field. Let $X$ and $Y$ be schemes over $k$ and let $\mathcal{F}$, resp. $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_X$-module, resp. $\mathcal{O}_Y$-module. Then we have a canonical isomorphism*

$$H^n(X \times_{\mathrm{Spec}(k)} Y, pr_1^*\mathcal{F} \otimes_{\mathcal{O}_{X\times_{\mathrm{Spec}(k)}Y}} pr_2^*\mathcal{G}) = \bigoplus\nolimits_{p+q=n} H^p(X,\mathcal{F}) \otimes_k H^q(Y,\mathcal{G})$$

*provided $X$ and $Y$ are quasi-compact and quasi-separated.*

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*Date*: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Proof.* If $X$ and $Y$ are separated or more generally have affine diagonal, then please see Lemma 0BED for "better" proof (the feature it has over this proof is that it identifies the maps as pullbacks followed by cup products). Let $X'$, resp. $Y'$ be the infinitesimal thickening of $X$, resp. $Y$ whose structure sheaf is $\mathcal{O}_{X'} = \mathcal{O}_X \oplus \mathcal{F}$, resp. $\mathcal{O}_{Y'} = \mathcal{O}_Y \oplus \mathcal{G}$ where $\mathcal{F}$, resp. $\mathcal{G}$ is an ideal of square zero. Then

$$\mathcal{O}_{X' \times Y'} = \mathcal{O}_{X \times Y} \oplus \mathrm{pr}_1^*\mathcal{F} \oplus \mathrm{pr}_2^*\mathcal{G} \oplus \mathrm{pr}_1^*\mathcal{F} \otimes_{\mathcal{O}_{X\times Y}} \mathrm{pr}_2^*\mathcal{G}$$

as sheaves on $X \times Y$. In this way we see that it suffices to prove that

$$H^n(X \times Y, \mathcal{O}_{X\times Y}) = \bigoplus\nolimits_{p+q=n} H^p(X, \mathcal{O}_X) \otimes_k H^q(Y, \mathcal{O}_Y)$$

for any pair of quasi-compact and quasi-separated schemes over $k$. Some details omitted.

To prove this statement we use cohomology and base change in the form of Cohomology of Schemes, Lemma 01XN. This lemma tells us there exists a bounded below complex of $k$-vector spaces, i.e., a complex $\mathcal{K}^\bullet$ of quasi-coherent modules on $\mathrm{Spec}(k)$, which universally computes the cohomology of $Y$ over $\mathrm{Spec}(k)$. In particular, we see that

$$R\mathrm{pr}_{1,*}(\mathcal{O}_{X\times Y}) \cong (X \to \mathrm{Spec}(k))^*\mathcal{K}^\bullet$$

in $D(\mathcal{O}_X)$. Up to homotopy the complex $\mathcal{K}^\bullet$ is isomorphic to $\bigoplus_{q\geq 0} H^q(Y, \mathcal{O}_Y)[-q]$ because this is true for every complex of vector spaces over a field. We conclude that

$$R\mathrm{pr}_{1,*}(\mathcal{O}_{X\times Y}) \cong \bigoplus\nolimits_{q\geq 0} H^q(Y, \mathcal{O}_Y)[-q] \otimes_k \mathcal{O}_X$$

in $D(\mathcal{O}_X)$. Then we have

$$\begin{aligned}
R\Gamma(X \times Y, \mathcal{O}_{X\times Y}) &= R\Gamma(X, R\mathrm{pr}_{1,*}(\mathcal{O}_{X\times Y})) \\
&= R\Gamma(X, \bigoplus\nolimits_{q\geq 0} H^q(Y, \mathcal{O}_Y)[-q] \otimes_k \mathcal{O}_X) \\
&= \bigoplus\nolimits_{q\geq 0} R\Gamma(X, H^q(Y, \mathcal{O}_Y) \otimes \mathcal{O}_X)[-q] \\
&= \bigoplus\nolimits_{q\geq 0} R\Gamma(X, \mathcal{O}_X) \otimes_k H^q(Y, \mathcal{O}_Y)[-q] \\
&= \bigoplus\nolimits_{p,q\geq 0} H^p(X, \mathcal{O}_X)[-p] \otimes_k H^q(Y, \mathcal{O}_Y)[-q]
\end{aligned}$$

as desired. The first equality by Leray for $\mathrm{pr}_1$ (Cohomology, Lemma 01EZ). The second by our decomposition of the total direct image given above. The third because cohomology always commutes with finite direct sums (and cohomology of $Y$ vanishes in sufficiently large degree by Cohomology of Schemes, Lemma 071L). The fourth because cohomology on $X$ commutes with infinite direct sums by Cohomology, Lemma 01FF. The final equality by our remark on the derived category of a field above. □